

[95]

SARDAR PATEL UNIVERSITY**M. Sc. Integrated Biotechnology (IBT) Semester 7 Examination - 2022****PS07EIBI22 – Microbial technology**

Date: 11/11/2022 (Friday)

Time: 10:00 am to 01:00 pm

Total Marks: 70**Note: (1) All the Questions are compulsory.****(2) Figures on the right indicate marks.****Q.1 Choose the correct option****1 × 8 = 08**

- (i) Prokaryotic microorganisms used in fermentation process _____.
[A] Protozoa [B] Fungi [C] Bacteria [D] All of these
- (ii) Microbial Alginate is _____ polymer
[A] An anionic [B] A cationic [C] A neutral [D] A basic
- (iii) Over production of Glutamic acid by *Corynebacterium glutamicum* can be achieved by _____.
[A] Addition of Glycine [C] Limiting concentration of Biotin
[B] Addition of Streptomycin [D] All of these
- (iv) Which of the following constituents of Milk are highly concentrated in Cheese?
[A] Lactose [B] Casein [C] Whey proteins [D] All of these
- (v) Which of the following is not a phytosterol?
[A] Sitosterol [B] Campesterol [C] Cholesterol [D] Stigmasterol
- (vi) The ICI process for SCP production was based on _____.
[A] n-paraffin [B] gas oil [C] methane [D] methanol
- (vii) Calf rennet is a/an _____.
[A] Milk protein [B] Lipase [C] Alkaline protease [D] Acid protease
- (viii) Which of the following is an inducer & Precursor for biosynthesis of ergot alkaloids?
[A] Glycine [B] Tryptophan [C] Mevalonic acid [D] Lysergic acid

Q-2 Attempt any seven of the following:**2 × 7 = 14**

- What is malolactic fermentation?
- Explain the effect of biotin on glutamic acid fermentation.
- Name the sterols used for synthesis of steroid drugs and write their sources.
- Explain the biological functions of microbial exopolysaccharides.
- Enlist the desirable characteristics of organism to be used as SCP.
- Enlist the applications of amylases.
- Explain the mode of action of Penicillin.
- Write the general characteristics of starter culture.
- Write the sources of Vitamin B₁₂.

- Q-3 [A] Explain in detail the biochemistry of citric acid overproduction and discuss the role of trace metal ions. [06]
[B] Explain in detail the fermentative production of amylase. [06]
OR
[B] Explain in detail the fermentative production of Glutamic acid. [06]
- Q-4 [A] Write on therapeutic use of steroids and explain any one microbial transformation of steroid in detail. [06]
[B] Explain the biosynthesis of Penicillin in detail. [06]
OR
[B] Write in detail on streptomycin recovery from fermentation broth. [06]
- Q-5 [A] Explain in detail important steps in cultivation of edible mushrooms. [06]
[B] Discuss various substrates used for the production of Bacterial SCP. [06]
OR
[B] Explain in detail on microbial production of PHA. [06]
- Q-6 [A] List and describe in detail the raw materials used in beer making. [06]
[B] Explain in detail microbiology & technology of Yoghurt manufacture. [06]
OR
[B] Explain in detail important steps in Wine manufacture. [06]

xxxxxx —X— xxxxxx.